# MCU-STM32

File: STM32G473.kicad_sch

## CLIFF

CLIFF_LM
CLIFF_RM
CLIFF_LED_EN
CLIFF_RR
CLIFF_RF
CLIFF_LR
CLIFF_LF

File: CLIFF.kicad_sch

## BUMPER

BUMP_L
BUMP_R
BUMP_L_LED_EN
BUMP_R_LED_EN

File: bumper_sensors.kicad_sch

## SIDE SENSORS

VSENS-3V3
SIDE-R1
ZX-SH1_0-7PWT
USART2-IR-R-RX
SIDE-PROXI-CE-R
I2C4_SCL
I2C4_SDA

## CARPET SENSOR

LO
ECHO_ADCD
CARPET-SENSOR-LOD
CARPET-SENSOR-ECHO

File: Carpet-sensor.kicad_sch

## MAIN BRUSH MOTOR

MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-ADC

File: MAIN-BRUSH-MOTOR .kicad_sch

## WATCHDOG

WDI
WD_OKD
INT-SENSORS
R2 47k

File: WATCHDOG.kicad_sch

## WHEEL MOTORs

WHEEL-MOT-R-IN2
WHEEL-MOT-R-IN1
WHEEL-MOT-R-ENCODED
WHEEL-MOT-R-DROPD
WHEEL-MOT-L-IN2
WHEEL-MOT-L-IN1
WHEEL-MOT-L-DROPD
WHEEL-MOT-L-ENCODED
WHEEL-MOT-R-ADCD
WHEEL-MOT-L-ADCD

File: WHEEL-MOTORs .kicad_sch

## LiDAR

WD_OK
LiDAR-RXD
LiDAR-TXD
LIDAR-MOTOR-CTRL
LiDAR-EN
LIDAR_RXD
LIDAR_TX
LIDAR-MOTOR-CTRLD

File: LIDAR.kicad_sch

## MAIN FAN

MAIN-FAN-FBD
MAIN-FAN-PWM

File: MAIN-FAN .kicad_sch

## USB-C POWER

DP
DN
USB_CC1D
USB_CC2
USB_VBUS_SENSE
USB_PWR_END
USB_CC1
USB_CC2
USB_VBUS_SENSE
USB_PWR_EN

File: usb_power.kicad_sch

## SIDE BRUSH MOTORs

SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-ADCD

File: SIDE-BRUSH-MOTORs .kicad_sch

## FRONT SENSORS

VSENS-3V3
VL-R1
2005FV-2X4P
R196 47k
VL-R-LPn
VL-L1
C6 10uF/10V
C19 10uF/10V
VL-I2C-RESETD
VL-I2C-RESET
UART3_RX2
IR-F-R
VL-L-LPnD
VL-L-LPn
UART3_RX1
IR-F-L
R6 2.2k
R5 2.2k

HALL-BIN1
HALL-TOWER1
ZX-SH1_0-3PLT
HALL-BIN
HALL-TOWER
REAR-IR1
REAR-IR-L
REAR-IR-R
ZX-SH1_0-4PLT

## IMU

IMU-INT#1D
IMU-SDA
IMU-SCL

File: IMU-ICM-4267-P .kicad_sch

## POWER CONTROL

VMOT-14V4-EN
STM-PWR-CTRL
PWR-BTN-SENSE
VSENS_3V3_EN
VSENS_5V_USB_EN
WD_OK
PWR-SW-G
POWER-BTN
BAT_ID
DC_IN_SENSE
BAT_SENSE

File: SYSTEM-POWER.kicad_sch

## BUTTON-LEDs

POWER-BTND
LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND

File: BUTTON-LEDs .kicad_sch

## BATTERY CHARGER

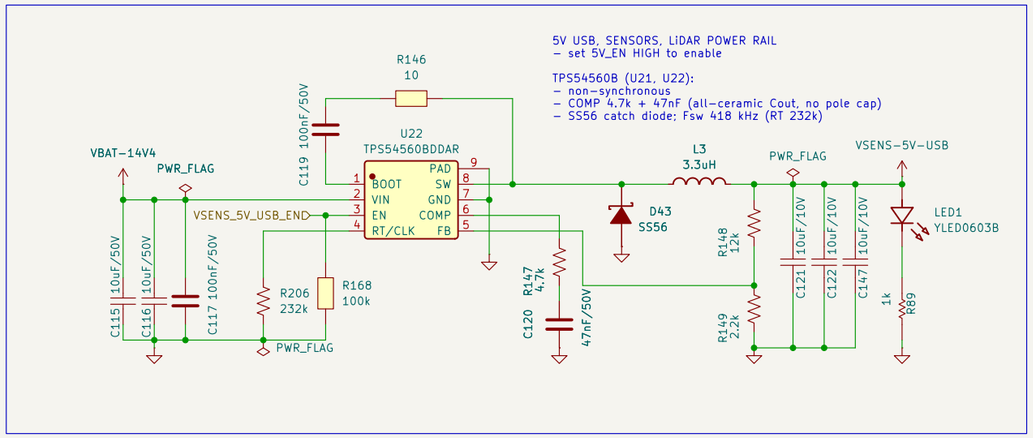
BATTERY-CHARGER
PWR-SW-G
BAT_SENSE
DC_IN_SENSE
ID

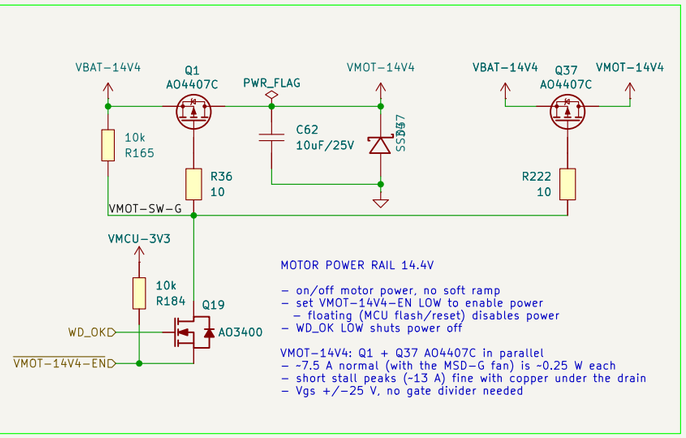
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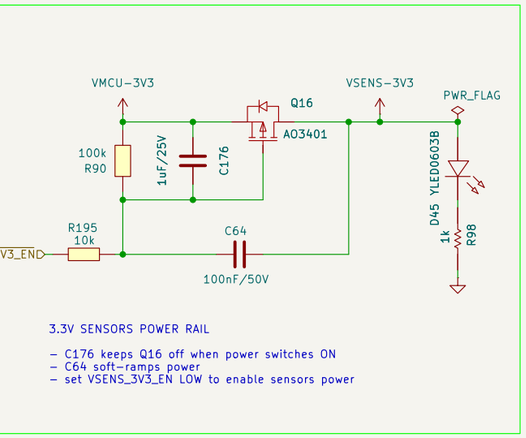
Sheet: /
File: Main.kicad_sch

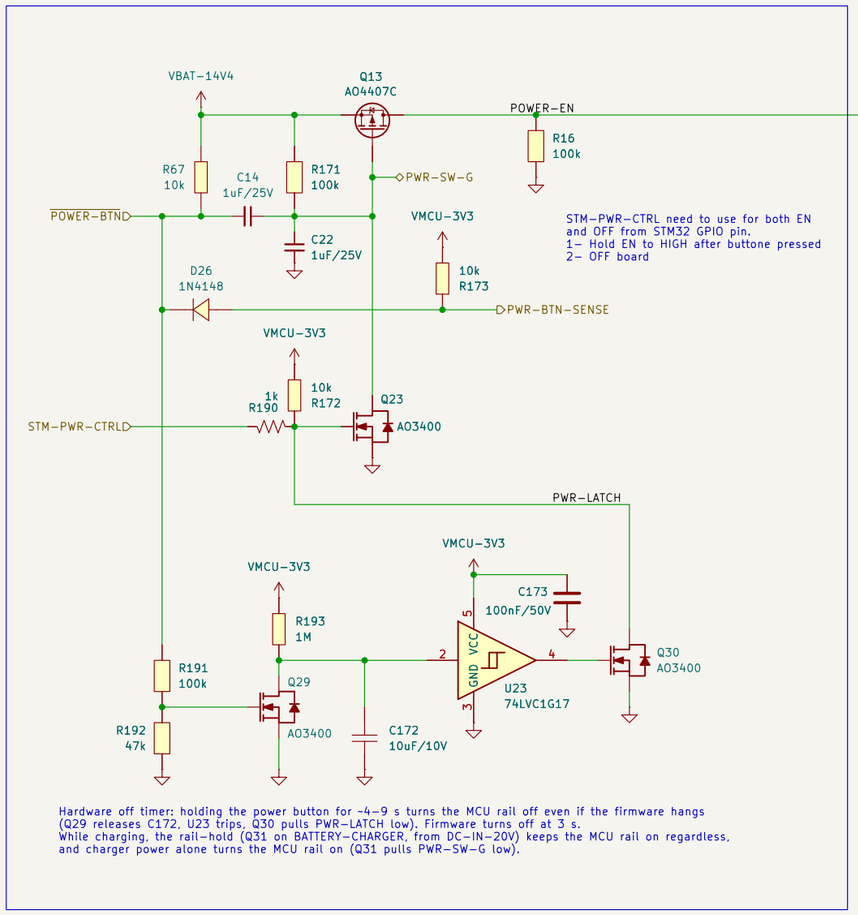
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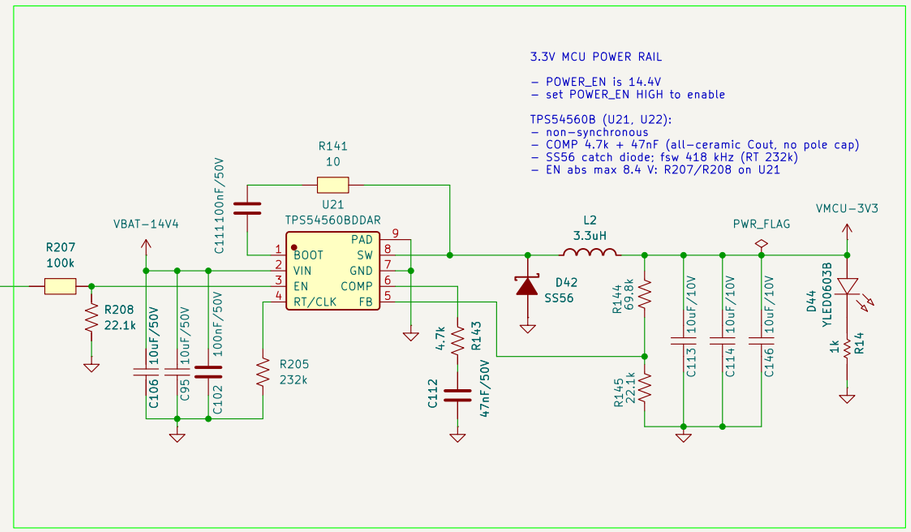
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| KiCad E.D.A. 10.0.4 | | Id: 1/16 |

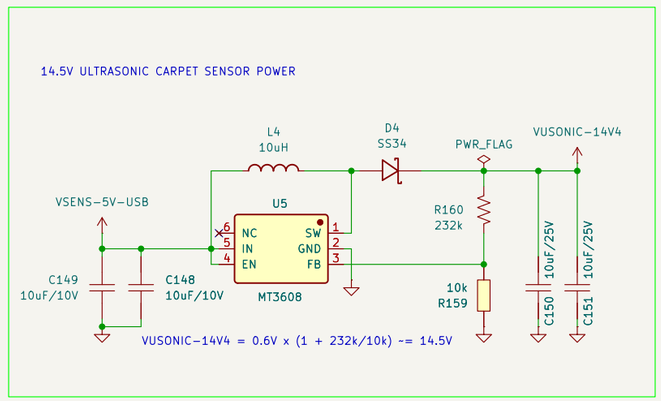


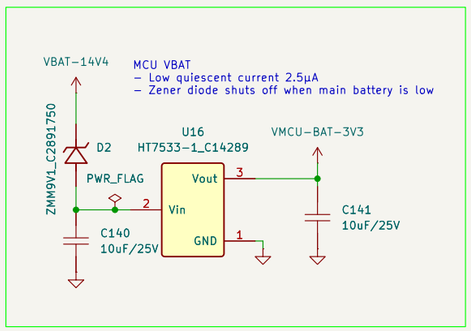



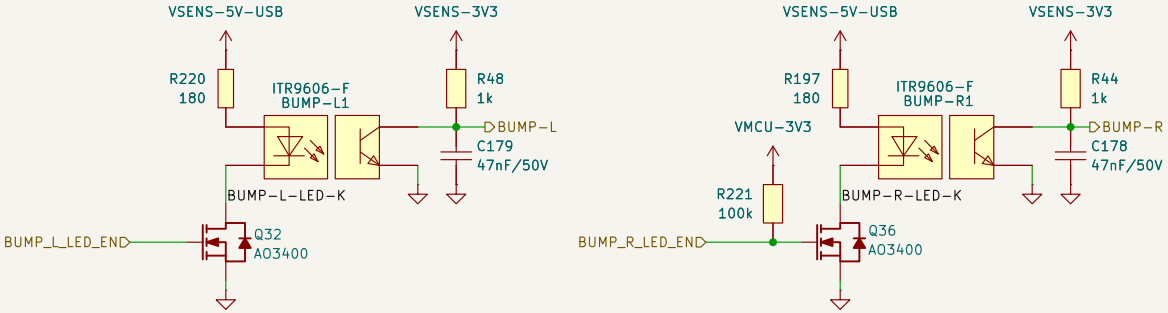



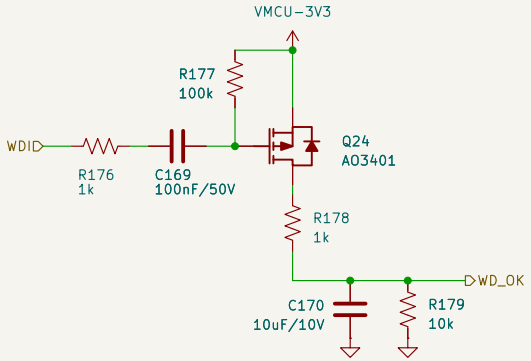

MP1 MP2 MP3 MP4 MP5 MP6

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|---|---|---|
| Sheet: /POWER/ | | |
| File: SYSTEM–POWER.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 2/16 |




- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range
  1k keeps photo transistor output out of saturation
  so PA4/PA5 (ADC2) see the transition (ITR9606 IC(ON) 0.5-10 mA)
- R221 enables MCU USB-C Rd on both CC1 and CC2 during MCU *reset*,
  so the phone detects the robot whichever way the cable is plugged in

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| Sheet: /Bumper Sensors/<br>File: bumper_sensors.kicad_sch | | |
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| KiCad E.D.A. 10.0.4 | | Id: 3/16 |




RC CHARGE-PUMP WATCHDOG

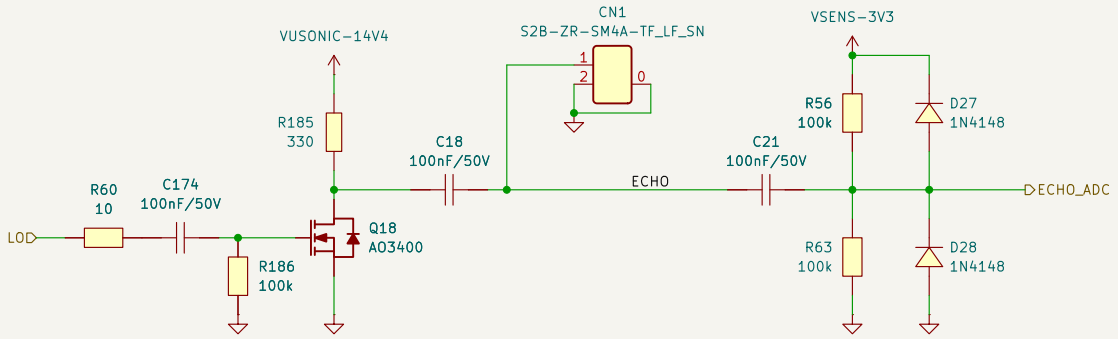
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

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| Sheet: /WATCHDOG/ | |
| File: WATCHDOG.kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
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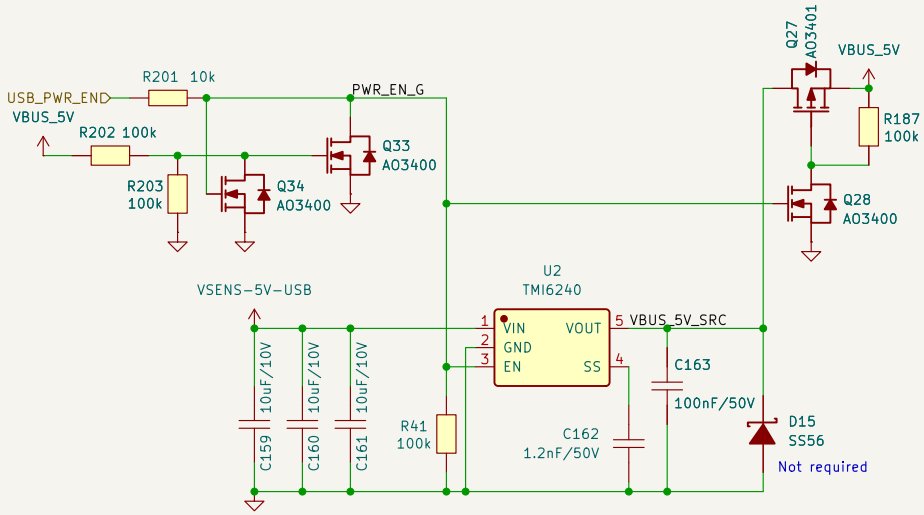
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



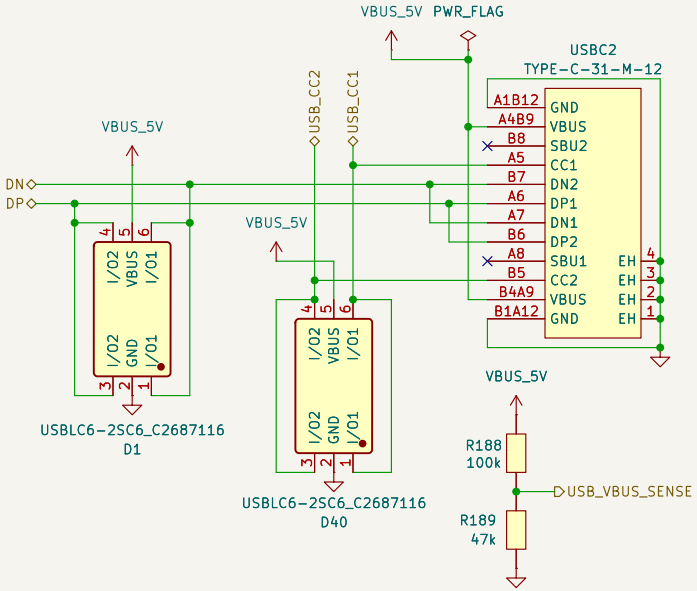

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|---|---|---|
| Sheet: /CARPET-SENSOR/ | | |
| File: Carpet-sensor.kicad_sch | | |
| **Title:** | | |
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| KiCad E.D.A. 10.0.4 | | Id: 6/16 |

# USB-C for SMARTPHONE



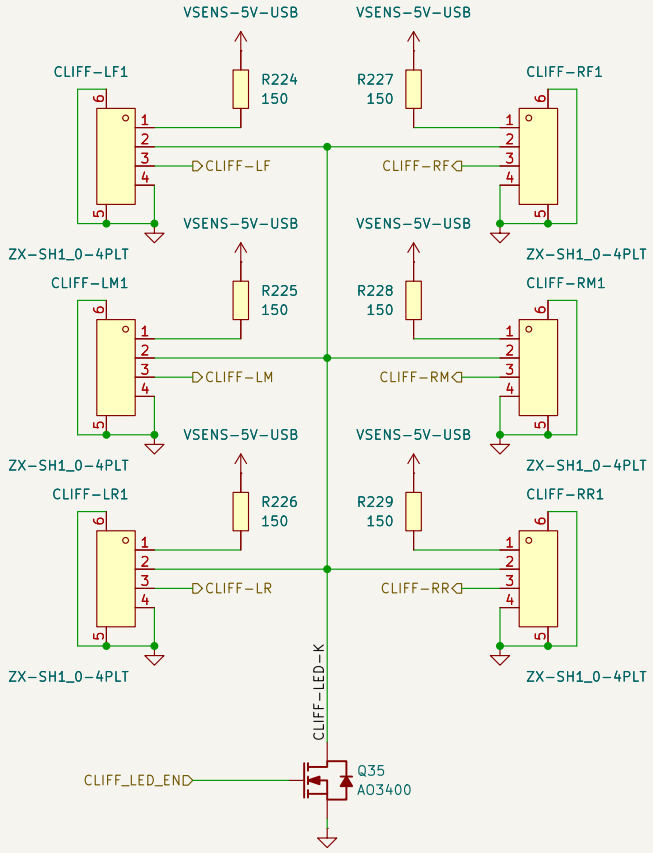

VBUS_5V lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS_5V is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS_5V; once on, Q34 latches it.

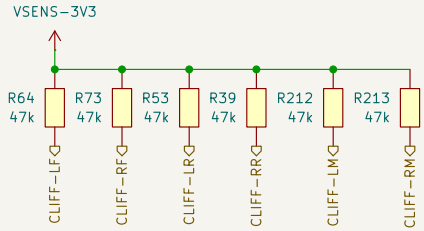
PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS_5V from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); USB_PWR_EN turns on TMI6240 and Q27 together.
TMI6240 now runs from VSENS-5V-USB (U25 buck removed).




USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default.

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| Sheet: /USB Power/ | |
| File: usb_power.kicad_sch | |
| **Title:** | |
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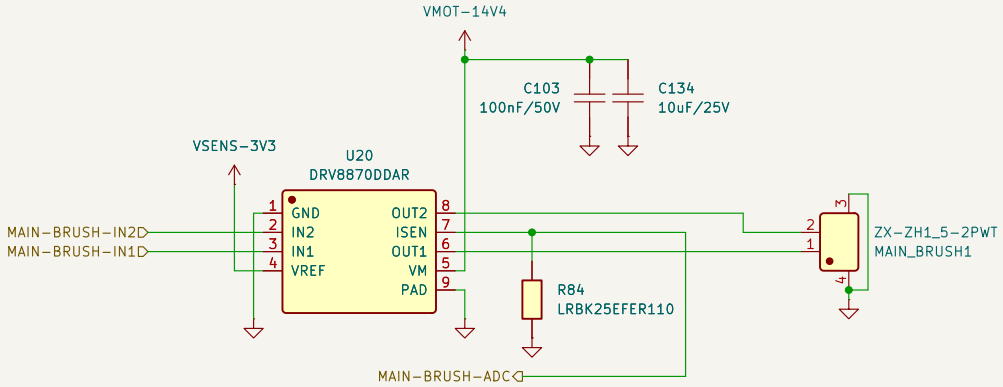




- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light
- possible TODO - cliff LEDs 2x or 3x in series, save resistors
  - worsens power rail noise rejection
  - ADC fw could reject noise by correlating the 6 readings
- 0.76W power consumption when enabled continuously
  - drive at ~10% duty cycle to save power
- tweak output pullup
  - a low resistor value can saturating the photo transistor
  - 47K pullup makes for 1ms output settling time
  - can use 10k for faster settling, to reduce duty cycle further
- combine low duty with fast duty LED cycling
  - fast (~10x photo transistor settling time) reduces LED average radiance to avoid photo transistor saturation
- check sensor faults - is output 3.3V or floating?
  - normal no-cliff condition - photo transisstor gets LED light
  Vary LED current and check photo-transistor voltage.

Sheet: /CLIFF/
File: CLIFF.kicad_sch

**Title:**

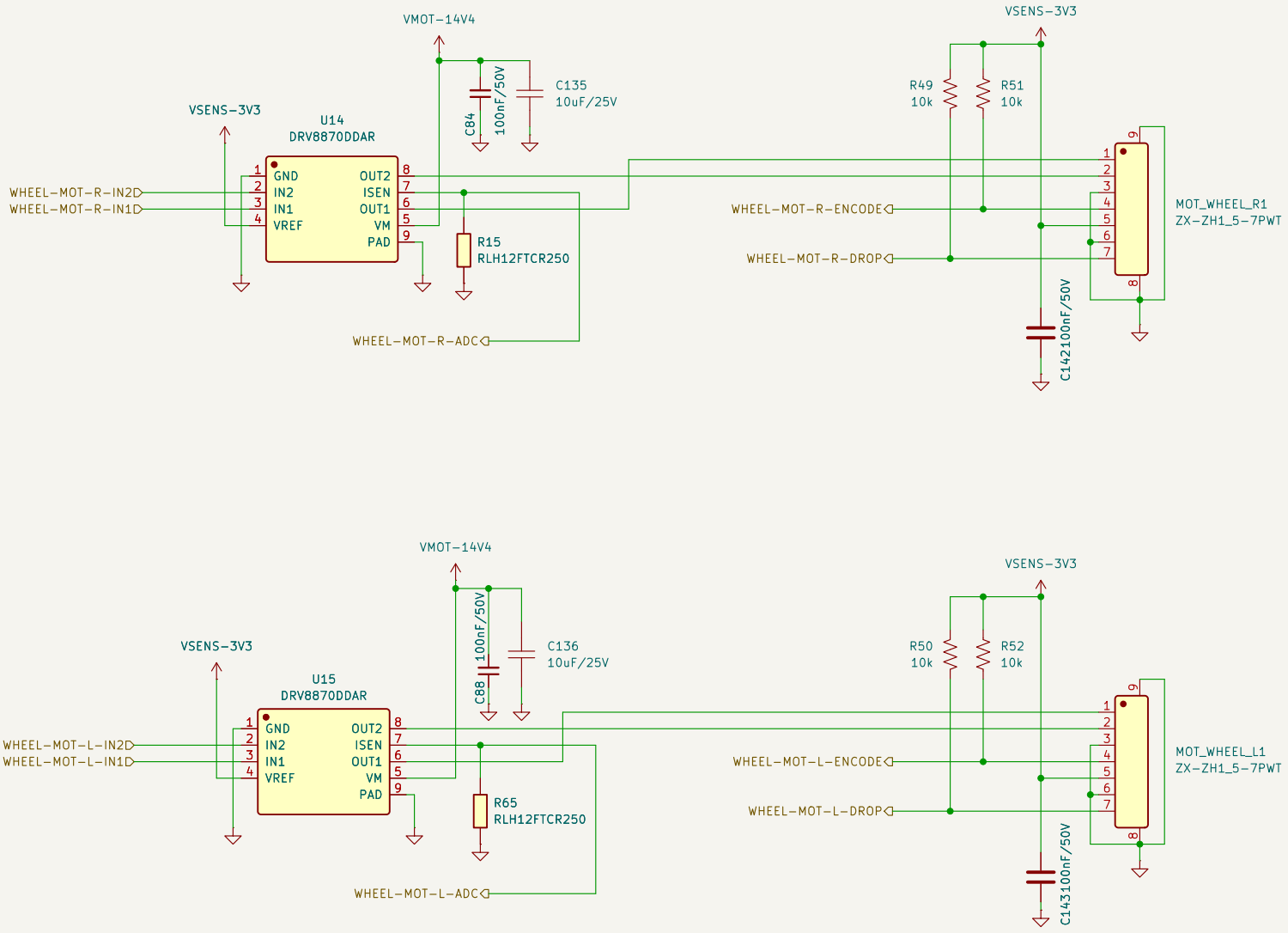
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KiCad E.D.A. 10.0.4 | Id: 19/16



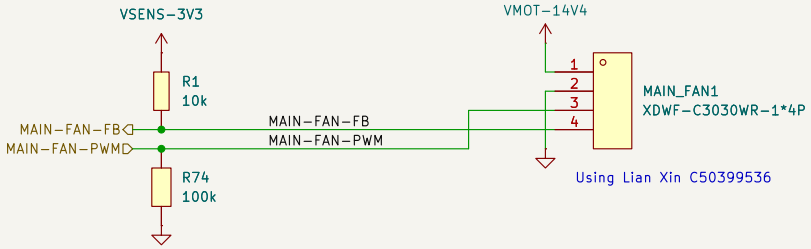

Sheet: /MAIN-BRUSH MOTOR/
File: MAIN-BRUSH-MOTOR .kicad_sch

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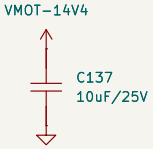
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KiCad E.D.A. 10.0.4 | Id: 22/16




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| Sheet: /WHEEL MOTOR/ | |
| File: WHEEL-MOTORs .kicad_sch | |
| **Title:** | |
| Size: USLetter | Date: |
| KiCad E.D.A. 10.0.4 | Id: 23/16 |



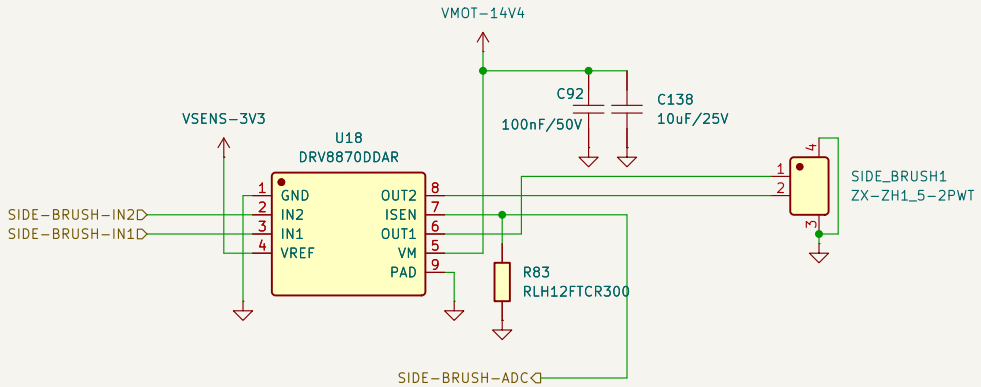

- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB



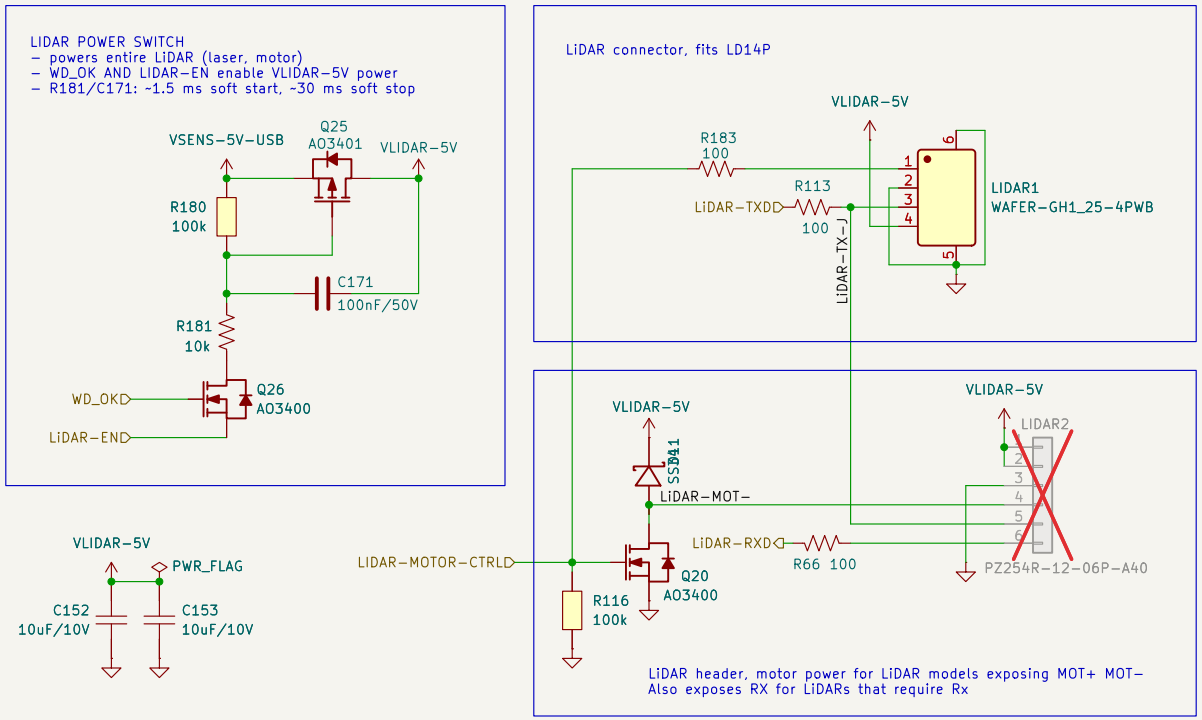

Sheet: /MAIN-FAN/
File: MAIN-FAN .kicad_sch

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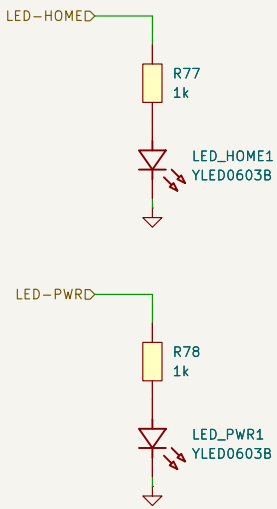
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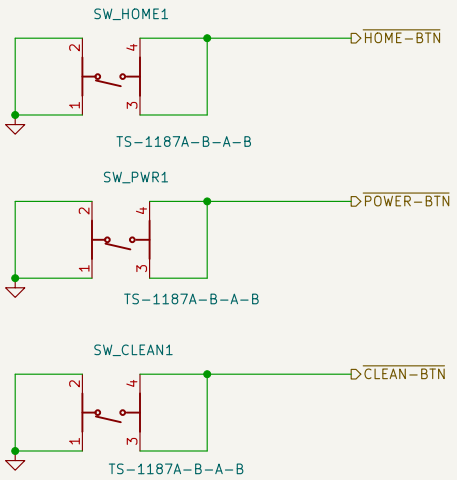



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| Sheet: /SIDE-BRUSH/ | |
| File: SIDE-BRUSH-MOTORs .kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 25/16 |

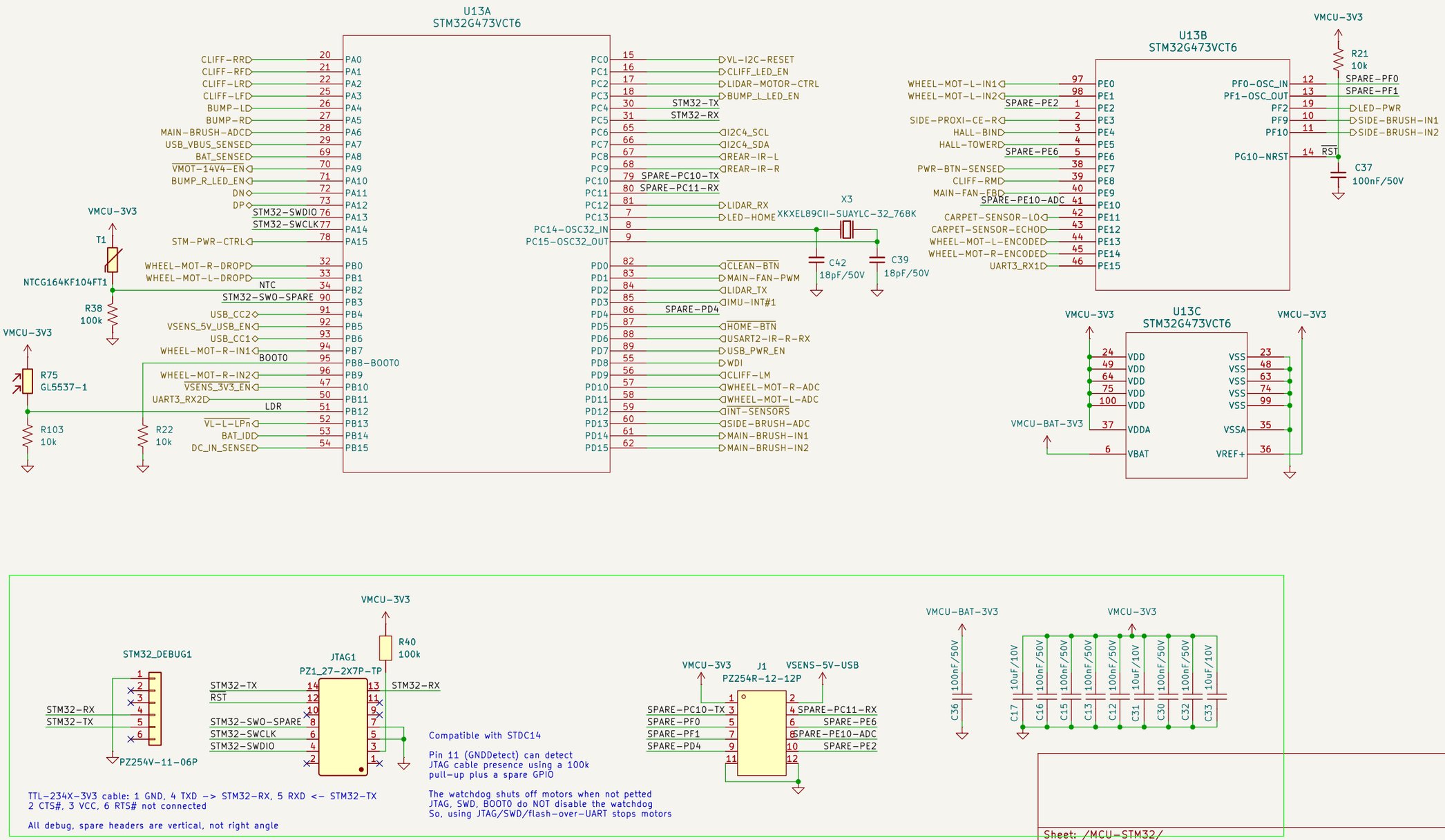


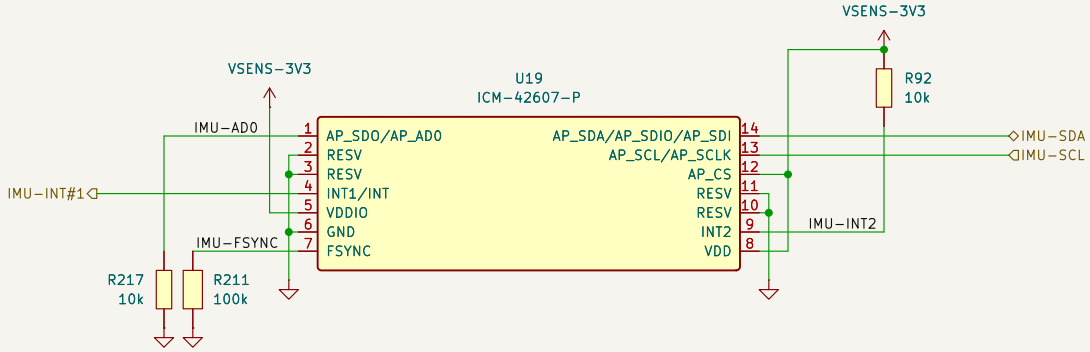

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| Sheet: /LIDAR/ | |
| File: LIDAR.kicad_sch | |
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| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 26/16 |



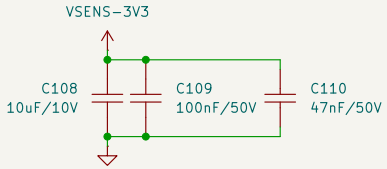




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| Sheet: /BUTTON-LED/ | |
| File: BUTTON-LEDs .kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 28/16 |


U13A
STM32G473VCT6
CLIFF-RR 20 PA0
CLIFF-RF 21 PA1
CLIFF-LR 22 PA2
CLIFF-LF 25 PA3
BUMP-LD 26 PA4
BUMP-RD 27 PA5
MAIN-BRUSH-ADC 28 PA6
USB_VBUS_SENSED 29 PA7
BAT_SENSED 69 PA8
VMOT-14V4-EN 70 PA9
BUMP_R_LED_EN 71 PA10
DN 72 PA11
DP 73 PA12
STM32-SWDIO 76 PA13
STM32-SWCLK 77 PA14
STM-PWR-CTRL 78 PA15
WHEEL-MOT-R-DROP 32 PB0
WHEEL-MOT-L-DROP 33 PB1
NTC 34 PB2
STM32-SWO-SPARE 90 PB3
USB_CC2 91 PB4
VSENS_5V_USB_EN 92 PB5
USB_CC1 93 PB6
WHEEL-MOT-R-IN1 94 PB7
BOOT0 95 PB8-BOOT0
WHEEL-MOT-R-IN2 96 PB9
VSENS_3V3_EN 47 PB10
UART3_RX2 50 PB11
LDR 51 PB12
VL-L-LPn 52 PB13
BAT_ID 53 PB14
DC_IN_SENSED 54 PB15
PC0 15 VL-I2C-RESET
PC1 16 CLIFF_LED_EN
PC2 17 LIDAR-MOTOR-CTRL
PC3 18 BUMP_L_LED_EN
PC4 30 STM32-TX
PC5 31 STM32-RX
PC6 65 I2C4_SCL
PC7 66 I2C4_SDA
PC8 67 REAR-IR-L
PC9 68 REAR-IR-R
PC10 79 SPARE-PC10-TX
PC11 80 SPARE-PC11-RX
PC12 81 LIDAR_RX
PC13 7 LED-HOME
PC14-OSC32_IN 8
PC15-OSC32_OUT 9
PD0 82 CLEAN-BTN
PD1 83 MAIN-FAN-PWM
PD2 84 LIDAR_TX
PD3 85 IMU-INT#1
PD4 86 SPARE-PD4
PD5 87 HOME-BTN
PD6 88 USART2-IR-R-RX
PD7 89 USB_PWR_EN
PD8 55 WDI
PD9 56 CLIFF-LM
PD10 57 WHEEL-MOT-R-ADC
PD11 58 WHEEL-MOT-L-ADC
PD12 59 INT-SENSORS
PD13 60 SIDE-BRUSH-ADC
PD14 61 MAIN-BRUSH-IN1
PD15 62 MAIN-BRUSH-IN2
X3
XKXEL89CII-SUAYLC-32_768K
C42 18pF/50V
C39 18pF/50V
VMCU-3V3
T1
NTCG164KF104FT1
R38 100k
R75 GL5537-1
R103 10k
R22 10k
U13B
STM32G473VCT6
WHEEL-MOT-L-IN1 97 PE0
WHEEL-MOT-L-IN2 98 PE1
SPARE-PE2 1 PE2
SIDE-PROXI-CE-R 2 PE3
HALL-BIN 3 PE4
HALL-TOWER 4 PE5
SPARE-PE6 5 PE6
PWR-BTN-SENSED 38 PE7
CLIFF-RM 39 PE8
MAIN-FAN-FB 40 PE9
SPARE-PE10-ADC 41 PE10
CARPET-SENSOR-LO 42 PE11
CARPET-SENSOR-ECHO 43 PE12
WHEEL-MOT-L-ENCODED 44 PE13
WHEEL-MOT-R-ENCODED 45 PE14
UART3_RX1 46 PE15
PF0-OSC_IN 12 SPARE-PF0
PF1-OSC_OUT 13 SPARE-PF1
PF2 19 LED-PWR
PF9 10 SIDE-BRUSH-IN1
PF10 11 SIDE-BRUSH-IN2
PG10-NRST 14 RST
R21 10k
C37 100nF/50V
U13C
STM32G473VCT6
VDD 24
VDD 49
VDD 64
VDD 75
VDD 100
VDDA 37
VBAT 6
VSS 23
VSS 48
VSS 63
VSS 74
VSS 99
VSSA 35
VREF+ 36
VMCU-BAT-3V3
STM32_DEBUG1
PZ254V-11-06P
JTAG1
PZ1_27-2X7P-TP
R40 100k
STM32-TX
RST
STM32-SWO-SPARE
STM32-SWCLK
STM32-SWDIO
STM32-RX
Compatible with STDC14
Pin 11 (GNDDetect) can detect JTAG cable presence using a 100k pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
All debug, spare headers are vertical, not right angle
J1
PZ254R-12-12P
SPARE-PC10-TX
SPARE-PF0
SPARE-PF1
SPARE-PD4
SPARE-PC11-RX
SPARE-PE6
SPARE-PE10-ADC
SPARE-PE2
VSENS-5V-USB
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch
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Rev:
Id: 30/16




IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

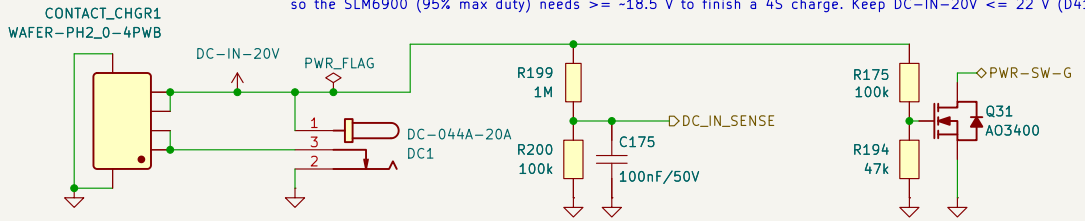



Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

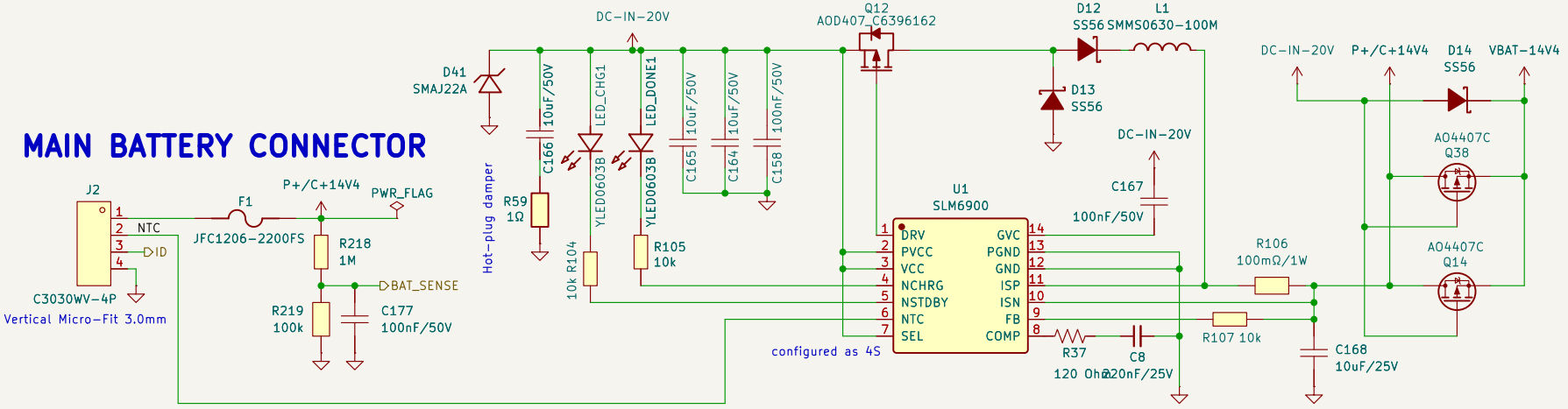
**Title:**

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KiCad E.D.A. 10.0.4 | Id: 32/16

DC-IN-20V: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT-14V4 = DC-IN-20V - 0.4 V,
so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC-IN-20V <= 22 V (D41, Q13 Vgs).




# MAIN BATTERY CONNECTOR




Rail-hold: charger power (dock or barrel jack)
turns the MCU rail on and keeps it on.
DC-IN-20V to GND ~130k (R175+R194 || R199+R200)
= dock presence signature.

R106 sets charging current
100 mOhm 1.2 A

| | |
|---|---|
| Sheet: /BATTERY-CHARGER/ | |
| File: Battery-Charger.kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 33/16 |